Universität St.Gallen

Institut für Computer Science
in Vorarlberg

Computer Networks

Network Security

Bruno Rodrigues

University of St. Gallen
Bachelor in Computer Science
Computer Networks (2,906)

From insight to impact

Lecture goals

- To **provide** a general overview of encryption mechanisms, approaches
- To **overview** the basics of symmetric (AES) and asymmetric (RSA) encryption, approaches, and their associated benefits and drawbacks
- To **comprehend** the concept of session keys (Diffie-Hellman) and Transport Layer Security (TLS)
- To **grasp** with the basics of certificates and digital signatures and their role ensuring a secure Web

Agenda

1. Encryption Mechanisms
2. Symmetric Encryption
3. Asymmetric Encryption
4. Session Keys and TLS
5. Certificates and Digital Signatures
6. Summary & Takeaways

Two-column content example

Symmetric encryption

- Single shared key
- Fast, suitable for bulk data
- Key distribution is the hard part

Asymmetric encryption

- Public / private key pair
- Solves key distribution
- Orders of magnitude slower

In practice, **TLS** combines both: asymmetric cryptography establishes a shared session key, which is then used with a symmetric cipher for the data stream.

Summary

- Symmetric cryptography is fast; asymmetric solves key distribution.
- **Certificates** bind public keys to identities; **TLS** uses them to bootstrap trust.
- Diffie-Hellman gives forward secrecy even if long-term keys leak later.

"

The cloud is just somebody else's computer.

Anonymous

Security is a process, not a product.

Questions?

University of St.Gallen

Prof. Dr. Bruno Rodrigues
bruno.rodrigues@unisg.ch

EQUIS ACCREDITED

AACSB ACCREDITED

ASSOCIATION OF MBAS ACCREDITED

"From insight to impact"